Infosys System Engineer: Hard-Level Mock Test — Paper 2

Total questions: 60 | **Duration:** 120 minutes | **Sectional cut-offs apply**

Section	Questions	Time
1. Reasoning Ability	Q1–15	25 min
2. Technical Ability (Quant)	Q16–25	35 min
3. Verbal Ability	Q26–45	20 min
4. Pseudo Code	Q46–50	10 min
5. Numeric Puzzle Solving	Q51–54	10 min
6. English Grammar	Q55–59	10 min
7. English Writing	Q60	10 min

SECTION 1: Reasoning Ability (15 Q, 25 min)

Q1. Statements: All servers are machines. Some machines are laptops. No laptop is portable.
Conclusions: I. Some machines are not portable. II. Some servers are laptops. III. No server is portable.
(a) Only I follows (b) I and III follow (c) I and II follow (d) None follows

Q2. Statements: All circles are squares. No square is a triangle. Some triangles are rectangles.
Conclusions: I. No circle is a triangle. II. Some rectangles are not squares. III. Some squares are not triangles.
(a) I and III only (b) I only (c) I, II and III (d) II and III only

Q3. *Data Sufficiency*. Is the integer n divisible by 6?
I. n is divisible by 4. II. n is divisible by 9.
(a) I alone is sufficient (b) II alone is sufficient (c) Both together are sufficient, neither alone (d) Even both together are not sufficient

Q4. *Data Sufficiency*. What is the cost price of an article?
I. It was sold at a profit of 20%. II. If sold at ₹150 more, the profit would have been 32%.
(a) I alone is sufficient (b) II alone is sufficient (c) Both together are sufficient, neither alone (d) Even both together are not sufficient

Q5. In a code, every letter is shifted 3 places forward in the alphabet (wrapping Z→C), so FLOWER is written as IORZHU. Using the same code, how is GARDEN written?
(a) JDUGHQ (b) JDUFHQ (c) JCUGHQ (d) JDVGHQ

Q6. In a code, "data flows fast" = "zu ka mo", "fast network grows" = "ka ta ri", "network data secure" = "zu ta ve". What is the code for "grows secure data"?
(a) ri ve zu (b) ta ve zu (c) ri ve ka (d) ka ve zu

Q7. A cube is painted red on one pair of opposite faces, blue on another pair, and green on the last pair. It is then cut into 27 identical small cubes. How many small cubes have exactly one face painted?
(a) 6 (b) 8 (c) 9 (d) 12

Q8. A rectangle is divided into a 3 × 3 grid of smaller cells by 2 evenly spaced vertical lines and 2 evenly spaced horizontal lines drawn inside it. How many rectangles (of any size) are there in the whole figure?
(a) 30 (b) 36 (c) 27 (d) 24

Q9. Pointing to a photograph, a man says, "She is the daughter of my grandfather's only son." How is the woman in the photograph related to the man?
(a) Sister (b) Daughter (c) Niece (d) Mother

Q10. If A ÷ B means A is the daughter of B, A + B means A is the son of B, and A − B means A is the wife of B, which expression correctly shows that M is the granddaughter of N?
(a) M ÷ P + N (b) M + P ÷ N (c) M − P ÷ N (d) M ÷ P − N

Q11. Priya starts at a point, faces south and walks 8 m. She turns left and walks 5 m, turns left again and walks 8 m, then turns right and walks 3 m. How far is she from the starting point, and in which direction?
(a) 8 m East (b) 8 m West (c) 2 m East (d) 2 m West

Q12. A man facing South turns 45° anticlockwise, then 90° clockwise, then 225° anticlockwise. Which direction is he facing now?
(a) North (b) South-East (c) South-West (d) West

Q13. Find the next term: 2, 6, 15, 31, 56, ?
(a) 92 (b) 88 (c) 90 (d) 84

Q14. Find the next term: 5, 11, 23, 47, 95, ?
(a) 189 (b) 191 (c) 193 (d) 190

Q15. Find the next letter-pair: CE, FI, IM, LQ, ?
(a) OT (b) OU (c) NT (d) PU

SECTION 2: Technical Ability, Quantitative (10 Q, 35 min)

Q16. In how many ways can the letters of the word MANAGEMENT be arranged so that all the vowels come together?
(a) 7560 (b) 15120 (c) 3780 (d) 30240

Q17. From a group of 6 men and 5 women, a team of 4 is to be selected such that it contains at most 2 women. In how many ways can this be done?
(a) 265 (b) 295 (c) 310 (d) 325

Q18. Three coins are tossed together, and it is known that at least one head appeared. What is the probability that exactly two heads appeared?
(a) 3/7 (b) 1/2 (c) 3/8 (d) 2/7

Q19. A box contains 6 defective and 14 non-defective bulbs. Two bulbs are drawn at random without replacement. What is the probability that at least one is defective?
(a) 99/190 (b) 91/190 (c) 33/95 (d) 29/95

Q20. A trader marks his goods at a price that would give him a 25% profit, but he actually gives a discount of 12% on the marked price and also uses a weight that is 10% less than it should be. Find his actual profit percentage.
(a) 20% (b) 22.2% (c) 25% (d) 18%

Q21. A and B together can complete a task in 15 days. They worked together for 5 days, after which A left and B completed the remaining work alone in 20 more days. In how many days can A alone complete the task?
(a) 20 days (b) 24 days (c) 25 days (d) 30 days

Q22. A boat's speed in still water is 15 km/h. It travels 60 km downstream and returns upstream to the starting point, taking a total of 9 hours. Find the speed of the stream.
(a) 4 km/h (b) 5 km/h (c) 6 km/h (d) 3 km/h

Q23. A town's population increases by 10% in the first year and decreases by 8% in the second year. If the population at the end of two years is 27,324, what was the population at the start?
(a) 27,000 (b) 26,500 (c) 26,000 (d) 27,500

Q24. A sum of money amounts to ₹13,000 after 3 years and to ₹15,000 after 5 years at the same rate of simple interest. Find the rate of interest per annum.
(a) 8% (b) 10% (c) 12% (d) 9%

Q25. ₹16,000 is invested for 2 years at compound interest, compounded annually, and yields the same interest as ₹18,000 invested for 2 years at 10% simple interest. Find the rate of compound interest (approximately).
(a) 9.5% (b) 10.5% (c) 11.5% (d) 12.5%

SECTION 3: Verbal Ability (20 Q, 20 min)

Read the passage and answer Q26–30.

> *Automation in enterprise software is frequently sold on the promise of eliminating human error, yet the more interesting effect is often a redistribution of error rather than its removal. When a system automates data entry, the mistakes that once occurred at the point of typing largely disappear, but new failure modes emerge upstream, in the design of the rules the automation follows, and downstream, in the operators' declining vigilance once they trust the system implicitly. A well-documented pattern in aviation and, increasingly, in IT operations is that highly reliable automation breeds complacency: engineers stop cross-checking outputs precisely because the system rarely errs, so on the rare occasion it does, the mistake propagates further before anyone notices. This does not argue against automation; it argues for designing monitoring and escalation processes that assume the automation will occasionally fail silently, rather than processes that assume a human will always catch what the machine misses.*

Q26. The passage primarily argues that:
(a) Automation should be avoided in critical systems
(b) Automation shifts the location of errors rather than eliminating them, so monitoring must assume silent failure
(c) Human operators are always more reliable than automated systems
(d) Aviation and IT face identical automation challenges

Q27. According to the passage, complacency arises because:
(a) operators are poorly trained
(b) the automation is rarely wrong, so vigilance declines
(c) the rules are too complex to follow
(d) engineers do not trust the system

Q28. The phrase "fail silently" in the passage most nearly means:
(a) the system stops working with an alarm
(b) the system errs without producing any visible warning
(c) the system asks for human confirmation
(d) the system shuts down automatically

Q29. Which of these would the author most likely recommend?
(a) Removing automation from enterprise systems entirely
(b) Designing escalation processes that do not rely solely on human vigilance
(c) Increasing the frequency of manual data entry
(d) Trusting automated systems completely once proven reliable

Q30. Which of the following is NOT stated or implied in the passage?
(a) New failure modes can appear in the design of automation rules
(b) Reliable automation can reduce the vigilance of human operators
(c) The passage argues automation should never be trusted
(d) Errors caught late can propagate further

Sentence Correction (Q31–35).

Q31. Each of the servers in the cluster **have** its own dedicated backup.
(a) No improvement (b) has (c) having (d) had

Q32. Scarcely had the deployment finished than the monitoring alerts started.
(a) No improvement (b) Scarcely had the deployment finished when (c) Scarcely did the deployment finish than (d) Scarcely the deployment had finished when

Q33. The report, along with its supporting data sheets, **were submitted** before the deadline.
(a) No improvement (b) was submitted (c) have been submitted (d) are submitted

Q34. If I was you, I would escalate this incident immediately.
(a) No improvement (b) If I were you (c) If I am you (d) If I had been you

Q35. Not only the developer but also the testers **was responsible** for the missed defect.
(a) No improvement (b) were responsible (c) is responsible (d) had been responsible

Para-jumbles (Q36–38).

Q36.
P: Yet within weeks, unexplained crashes began appearing in production.
Q: The new caching layer initially seemed to improve response times dramatically.
R: Investigation traced the crashes to a race condition the load tests had never triggered.
S: The team subsequently rewrote the load-testing suite to include concurrent write scenarios.
(a) QPRS (b) QRPS (c) PQRS (d) QSPR

Q37.
P: They are, instead, thought experiments that isolate one variable at a time.
Q: Economic models are often criticised for their unrealistic assumptions.
R: This deliberate simplification is what makes them useful, not despite the unrealism but because of it.
S: Critics forget that the models were never meant to be literal descriptions of the economy.
(a) QSPR (b) QPSR (c) QSRP (d) SQPR

Q38.
P: Consequently, the interface today handles ten times the daily traffic it was designed for.
Q: The original API was built to serve a handful of internal tools.
R: Over time, external partners began integrating with it as well.
S: No one anticipated that this informal adoption would continue for years.
(a) QRSP (b) QSRP (c) QRPS (d) RQSP

Vocabulary, Synonyms and Antonyms (Q39–45).

Q39. The consultant's **cursory** review missed several critical vulnerabilities. "Cursory" means:
(a) thorough (b) hasty and superficial (c) expensive (d) delayed

Q40. The new policy was met with **tacit** approval from the board. "Tacit" means:
(a) loudly announced (b) implied without being stated (c) formally written (d) rejected

Q41. Fill in the blank: Despite the outage, the support team remained _____, calmly working through the incident.

(a) frantic (b) unflappable (c) indifferent (d) negligent

Q42. Synonym of **PRAGMATIC**:

(a) idealistic (b) practical (c) theoretical (d) emotional

Q43. Synonym of **VOLATILE**:

(a) stable (b) unpredictable (c) calm (d) gradual

Q44. Antonym of **CONCISE**:

(a) brief (b) verbose (c) clear (d) precise

Q45. Antonym of **DILIGENT**:

(a) hardworking (b) careful (c) lazy (d) thorough

SECTION 4: Pseudo Code (5 Q, 10 min)

Q46. What is printed?

```
int f(int n) {
    if (n == 0) return 1;
    if (n == 1) return 2;
    return f(n-1) + f(n-2) * 2;
}
print(f(5))
```

(a) 26 (b) 30 (c) 32 (d) 34

Q47. What is printed? (x and y are unsigned integers)

```
x = 58; y = 0;
while (x > 0) {
    y = (y << 1) | (x & 1);
    x = x >> 1;
}
print(y)
```

(a) 23 (b) 29 (c) 37 (d) 41

Q48. What is printed?

```
s = 0
for i = 1 to 4
    for j = 1 to 4
        if (i == j) continue
        if (i + j > 6) break
        s = s + (i * j)
    end for
end for
print(s)
```

(a) 40 (b) 44 (c) 46 (d) 48

Q49. A queue is used (FIFO). What is printed?

```
enqueue(2); enqueue(4); enqueue(6);
a = dequeue();
enqueue(a + 10);
b = dequeue();
c = dequeue();
print(b + c)
```

(a) 10 (b) 16 (c) 18 (d) 22

Q50. What is the final content of `arr`? (0-indexed; each statement uses the array's state at that point)

```
arr = [1, 2, 3, 4, 5]
temp = arr[0]
arr[0] = arr[4]
arr[4] = arr[2]
arr[2] = arr[1]
arr[1] = temp
```

(a) [5, 1, 2, 4, 3] (b) [5, 1, 3, 4, 2] (c) [5, 2, 1, 4, 3] (d) [5, 1, 2, 4, 2]

SECTION 5: Numeric Puzzle Solving (4 Q, 10 min)

Q51. *Matchstick.* Five identical unit squares are arranged in a row sharing sides, using 16 matchsticks in total. What is the minimum number of sticks that must be removed so that exactly 3 complete unit squares remain?

(a) 1 (b) 2 (c) 3 (d) 4

Q52. *Circular wheel.* Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.

Wheel 1: T = 6, R = 2, B = 4, L = 3, Centre = 30

Wheel 2: T = 5, R = 3, B = 8, L = 2, Centre = 46

Wheel 3: T = 7, R = 4, B = 3, L = 6, Centre = ?

(a) 39 (b) 41 (c) 45 (d) 49

Q53. *Matrix logic.* Find the missing number.

2	3	11
3	4	19
4	5	?

(a) 27 (b) 29 (c) 31 (d) 33

Q54. *Pattern.* Find the next term: 1, 4, 10, 20, 35, ?

(a) 52 (b) 56 (c) 60 (d) 49

SECTION 6: English Grammar (5 Q, 10 min)

Q55. Each of the following options _____ its own trade-offs, so the team must weigh them carefully.

(a) have (b) has (c) having (d) were having

Q56. By next Friday, the QA team _____ all the regression test cases for this release.

(a) finished (b) will finish (c) will have finished (d) finishes

Q57. The architect is known _____ his attention to detail but is often impatient _____ incomplete requirements.

(a) for, with (b) of, by (c) for, of (d) by, for

Q58. *Error spotting.* "The manager, together with his team members, (A) / are reviewing the proposal (B) / that was submitted (C) / earlier this week. (D)"

(a) A (b) B (c) C (d) D

Q59. He bought ___ umbrella, ___ one-way ticket, and ___ useful travel guide before ___ trip.

(a) an, a, a, the (b) a, a, an, the (c) an, a, a, a (d) a, an, a, the

SECTION 7: English Writing (1 Q, 10 min)

Q60. *Write an email of 120–150 words.*

You are a System Engineer supporting a client's application in production. You have identified that a recurring performance issue is caused by an outdated server configuration, and a fix requires a scheduled maintenance window that will cause 30 minutes of downtime. Write an email to the client's IT coordinator explaining the issue, proposing a suitable maintenance window, describing the expected impact, and requesting their approval to proceed.

Evaluation rubric (10 marks): Subject line and format (2) | Clarity and completeness of content (3) | Professional tone and courtesy (2) | Grammar, spelling and vocabulary (2) | Word limit adherence (1)

ANSWER KEY WITH EXPLANATIONS

Section 1: Reasoning

1. **(a)** Some machines are laptops, and no laptop is portable, so those machines are not portable — I follows. Nothing links servers specifically to laptops or to portability, so II and III don't follow.

2. **(c)** Circles ⊆ squares, and squares have no overlap with triangles, so circles have none either (I). Some triangles are rectangles, and since no triangle is a square, those rectangles are not squares (II). Since no square is a triangle, it's also true that some squares are not triangles (III).

3. **(c)** I alone fails (n = 4 is not divisible by 6). II alone fails (n = 9 is not divisible by 6). Together, a number divisible by both 4 and 9 is divisible by their LCM, 36, which is always divisible by 6.

4. **(b)** II alone: 12% of CP = ₹150, so CP = ₹1,250 — sufficient. I alone gives only a percentage, not an absolute value.

5. **(a)** Applying the same +3 shift: G→J, A→D, R→U, D→G, E→H, N→Q, giving **JDUGHQ**.

6. **(a)** "fast" is common to statements 1 and 2 → ka. "data" is common to 1 and 3 → zu. "network" is common to 2 and 3 → ta. By elimination: flows = mo, grows = ri, secure = ve. So "grows secure data" = **ri ve zu**.

7. **(a)** In a 3×3×3 cube, only the centre cube of each face touches just one painted face — 1 per face × 6 faces = **6**, regardless of the colour scheme.

8. **(b)** With 2 interior + 2 boundary lines in each direction, there are 4 vertical and 4 horizontal lines. Rectangles = C(4,2) × C(4,2) = 6 × 6 = **36**.

9. **(a)** The grandfather's only son is the man's father. The daughter of the man's father is his **sister**.

10. **(a)** M ÷ P: M is daughter of P. P + N: P is son of N. So M is P's daughter, and P is N's son — M is N's granddaughter.

11. **(a)** Tracking positions from (0,0): south 8 → (0,−8); left turn faces east, walk 5 → (5,−8); left turn faces north, walk 8 → (5,0); right turn faces east, walk 3 → (8,0). Net displacement is **8 m East**.

12. **(a)** Using bearings (clockwise from North): South = 180°. −45° (anticlockwise) = 135°; +90° (clockwise) = 225°; −225° (anticlockwise) = 0°, which is **North**.

13. **(a)** Differences are 4, 9, 16, 25 (perfect squares), so the next difference is 36: 56 + 36 = **92**.

14. **(b)** Each term = previous × 2 + 1: 95 × 2 + 1 = **191**.

15. **(b)** First letters C, F, I, L rise by 3 each time → next is O. Second letters E, I, M, Q rise by 4 each time → next is U. Answer: **OU**.

Section 2: Quant

16. **(a)** MANAGEMENT has M2, A2, N2, G1, E2, T1 (10 letters). Vowels A,A,E,E form one block: internal arrangements = 4!/(2!2!) = 6. Treating the block as one unit alongside the 6 consonants (M2,N2,G1,T1) gives 7 units: 7!/(2!2!) = 1260. Total = 1260 × 6 = **7,560**.

17. **(a)** 0 women: C(6,4)=15. 1 woman: C(6,3)×C(5,1)=100. 2 women: C(6,2)×C(5,2)=150. Total = 15+100+150 = **265**.

18. **(a)** Excluding TTT leaves 7 equally likely outcomes. Exactly two heads: HHT, HTH, THH = 3 outcomes. Probability = **3/7**.

19. **(a)** P(no defective) = (14×13)/(20×19) = 91/190. P(at least one defective) = 1 − 91/190 = **99/190**.

20. **(b)** Take CP = 100 per unit weight, so marked price = 125 (25% intended profit). After a 12% discount, SP = 125 × 0.88 = 110. But he delivers only 0.9 units (10% short weight) whose true cost is 90. Profit = 110 − 90 = 20 on a cost of 90, giving 20/90 × 100 ≈ **22.2%**.

21. **(d)** In 5 days together, they finish 5/15 = 1/3. The remaining 2/3 is done by B alone in 20 days, so B's rate = (2/3)/20 = 1/30/day (B alone takes 30 days). Combined rate = 1/15, so A's rate = 1/15 − 1/30 = 1/30/day — A alone also takes **30 days**.

22. **(b)** 60/(15+s) + 60/(15−s) = 9. Simplifying: 1800 = 9(225 − s^2), so s^2 = 25, s = **5 km/h**.

23. **(a)** P × 1.10 × 0.92 = 27,324, so P × 1.012 = 27,324, giving P = **27,000**.

24. **(b)** Interest for 2 years (year 5 − year 3) = 15,000 − 13,000 = 2,000, so 1 year's interest = 1,000. Principal = 13,000 − 3×1,000 = 10,000. Rate = 1,000/10,000 × 100 = **10%**.

25. **(b)** SI on ₹18,000 at 10% for 2 years = ₹3,600. Setting CI on ₹16,000 equal: 16,000×((1+r)² − 1) = 3,600 gives (1+r)² = 1.225, so r ≈ **10.5%**.

Section 3: Verbal

26. **(b)** 27. **(b)** 28. **(b)** 29. **(b)** 30. **(c)**

31. **(b)** "Each" is singular, so the verb must be "has".

32. **(b)** "Scarcely…when" is the correct pairing, not "than".

33. **(b)** "The report" is the true (singular) subject; "along with…" does not pluralize it, so "was submitted".

34. **(b)** Hypothetical conditions use the subjunctive "were" after "if", regardless of subject.

35. **(b)** With "not only…but also" joining two subjects, the verb agrees with the nearer one (testers, plural), so "were responsible".

36. **(a)** Q→P→R→S. 37. **(a)** Q→S→P→R. 38. **(a)** Q→R→S→P.

39. **(b)** 40. **(b)** 41. **(b)** 42. **(b)** 43. **(b)** 44. **(b)** 45. **(c)**

Section 4: Pseudo Code

46. **(c) 32.** f(0)=1, f(1)=2, f(2)=2+2=4, f(3)=4+4=8, f(4)=8+8=16, f(5)=16+16=**32**.

47. **(a) 23.** The loop peels off bits of x from the LSB and rebuilds y by shifting left each time, which reverses the bit order of x within its significant bits. 58 = 111010_2 (6 bits); reversed = 010111_2 = **23**. (Trace: y goes 0→0→1→2→5→11→23 as x's bits 0,1,0,1,1,1 are consumed in order.)

48. **(c) 46.** i=1: skips j=1 (i==j); adds 1×2+1×3+1×4 = 9. i=2: skips j=2; adds 2×1+2×3+2×4 = 16 (running 25; note j=4 gives i+j=6, which is not >6, so it's included). i=3: skips j=3; adds 3×1+3×2 = 9, then j=4 gives i+j=7>6 → break (running 34). i=4: adds 4×1+4×2 = 12, then j=3 gives i+j=7>6 → break (running **46**).

49. **(a) 10.** Queue [2,4,6] → dequeue 2 (a) → [4,6] → enqueue 12 → [4,6,12] → dequeue 4 (b) → dequeue 6 (c) → [12]. b + c = 4 + 6 = **10**.

50. **(a)** temp=1. arr[0]=arr[4]=5 → [5,2,3,4,5]. arr[4]=arr[2]=3 → [5,2,3,4,3]. arr[2]=arr[1]=2 → [5,2,2,4,3]. arr[1]=temp=1 → final **[5,1,2,4,3]**.

Section 5: Numeric Puzzles

51. **(a) 1.** The stick shared between the 2nd and 3rd squares is a side of both. Removing that single stick breaks both squares at once, leaving the 1st, 4th and 5th squares intact — exactly 3 complete squares, with only 1 stick removed.

52. **(c) 45.** Rule: Centre = (T × B) + (R × L). Wheel 1: (6×4)+(2×3) = 24+6 = 30 ✓. Wheel 2: (5×8)+(3×2) = 40+6 = 46 ✓. Wheel 3: (7×3)+(4×6) = 21+24 = **45**.

53. **(b) 29.** Rule: third column = (a×b) + a + b. Row 1: (2×3)+2+3 = 11 ✓. Row 2: (3×4)+3+4 = 19 ✓. Row 3: (4×5)+4+5 = 20+9 = **29**.

54. **(b) 56.** These are tetrahedral numbers, C(n+2,3): 1, 4, 10, 20, 35, and next is C(8,3) = **56**.

Section 6: Grammar

55. **(b)** "Each" is singular → has.

56. **(c)** Future perfect ("will have finished") for an action complete before a future point.

57. **(a)** known **for**, impatient **with**.

58. **(b)** The true subject is "The manager" (singular); "together with his team members" doesn't pluralize it, so it should be "is reviewing" — the error is in part (B).

59. **(a)** an umbrella (vowel sound), a one-way ticket ("w" sound), a useful travel guide ("y" sound), the trip (specific, already referenced).

Section 7: Writing

Use the rubric above. Model structure: Subject line ("Scheduled Maintenance Window for Server Configuration Fix"), greeting, a brief statement of the issue (outdated configuration causing recurring performance degradation), the proposed maintenance window and expected 30-minute downtime, the mitigation/communication plan, a request for approval, and a courteous sign-off with name and designation.

Scoring guide

Award +1 per MCQ (Q1–59) and up to 10 for Q60. Aim to clear each section's cut-off separately, not just the total.